

AePW-4 High-Angle Working Group: An Overview of Recent Progress and Future Directions

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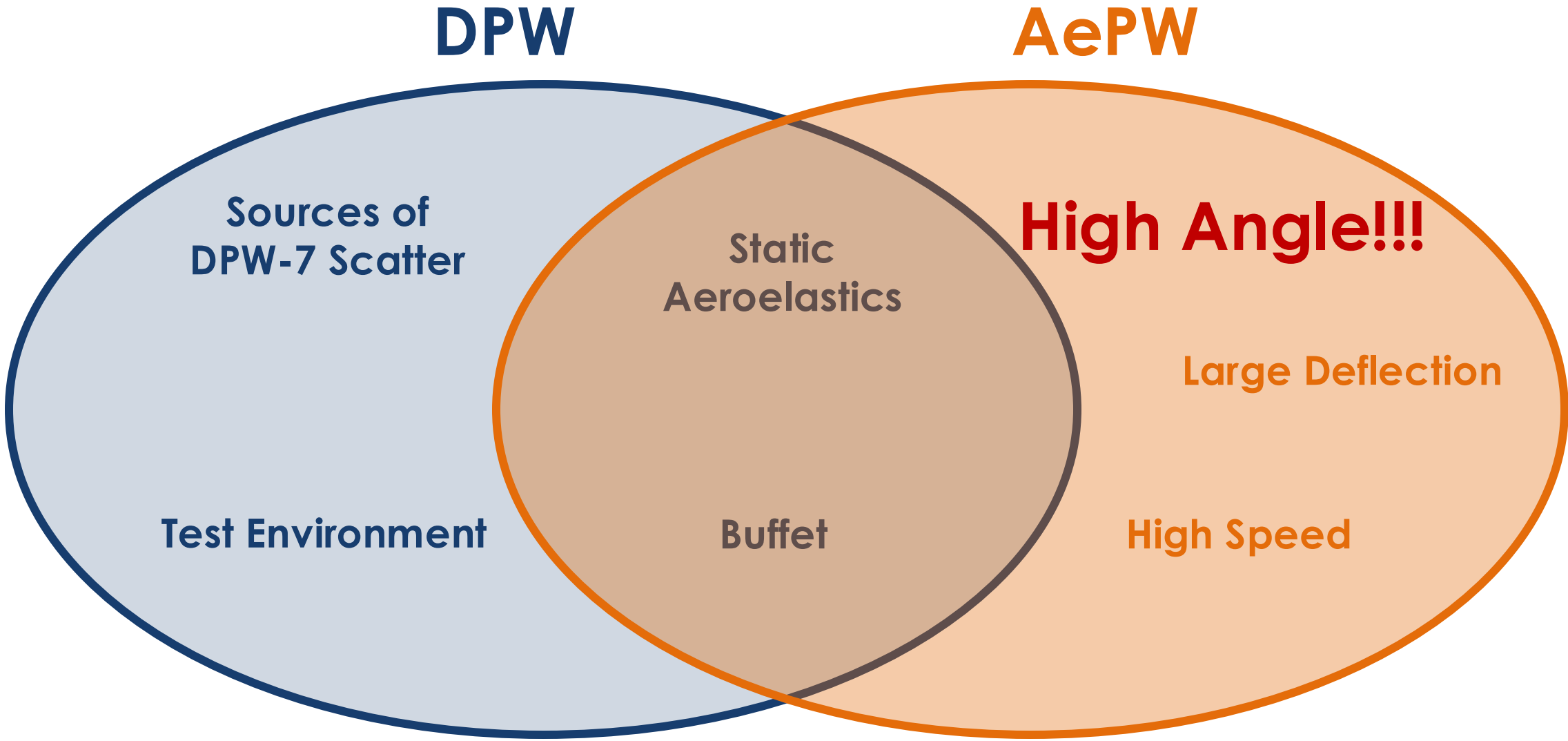


Presented by
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[https://nescacademy.nasa.gov/
workshops/AePW4/public](https://nescacademy.nasa.gov/workshops/AePW4/public)

<https://aiaa-dpw.larc.nasa.gov>



Benchmark SuperCritical Wing (BSCW) Wing Configuration Past Workshop Conditions

- **AePW-1 (2012):**
 - Steady-rigid and forced-oscillation cases at Mach 0.85, AoA = 5° ✓
- **AePW-2 (2016):**
 - Forced-oscillation case at Mach 0.70, AoA = 3° ✓
 - Flutter prediction at Mach 0.74, AoA = 0° ✓
 - Unsteady-rigid, forced-oscillation, and flutter cases at Mach 0.85, 5° ✓ ✓ ✓
- **AePW-3 (2023):**
 - Flutter prediction at Mach 0.80, AoA = 5° ✓
 - Shock-buffet case at Mach 0.80, AoA = 5° ✓

✓ - good

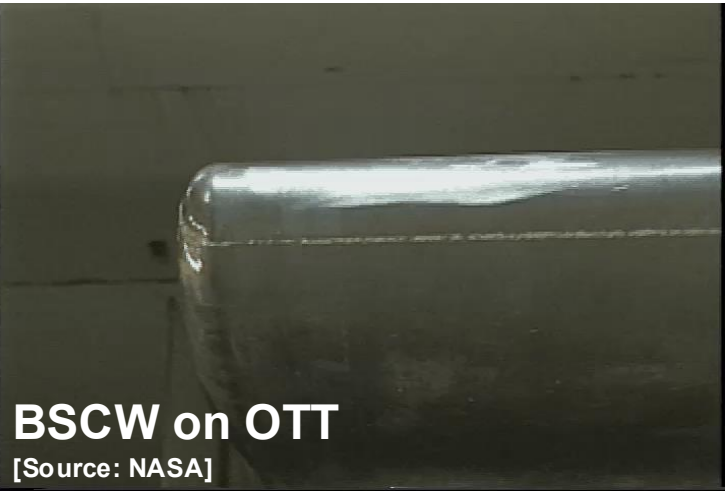
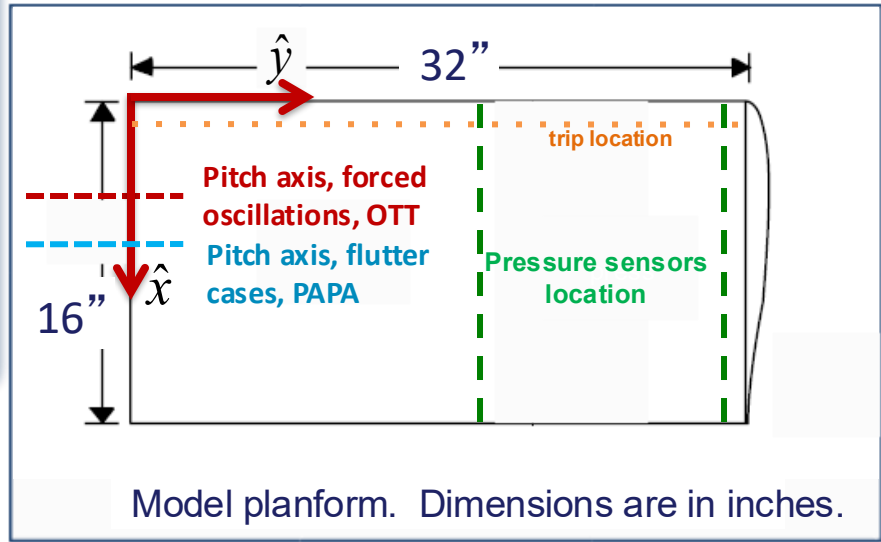
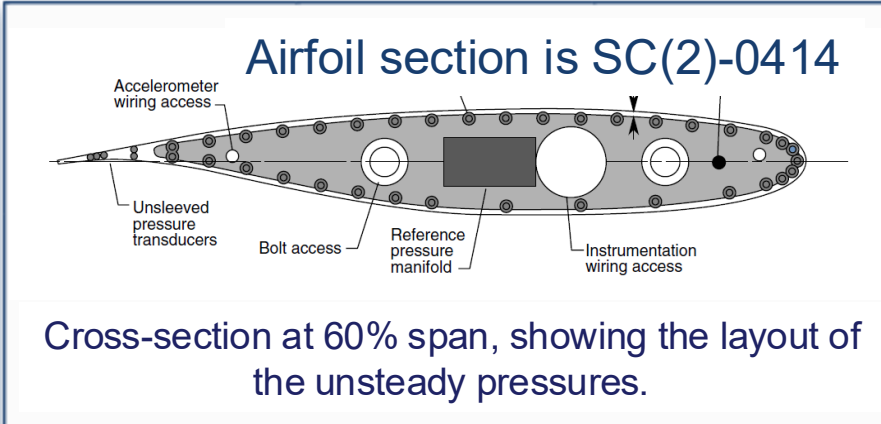
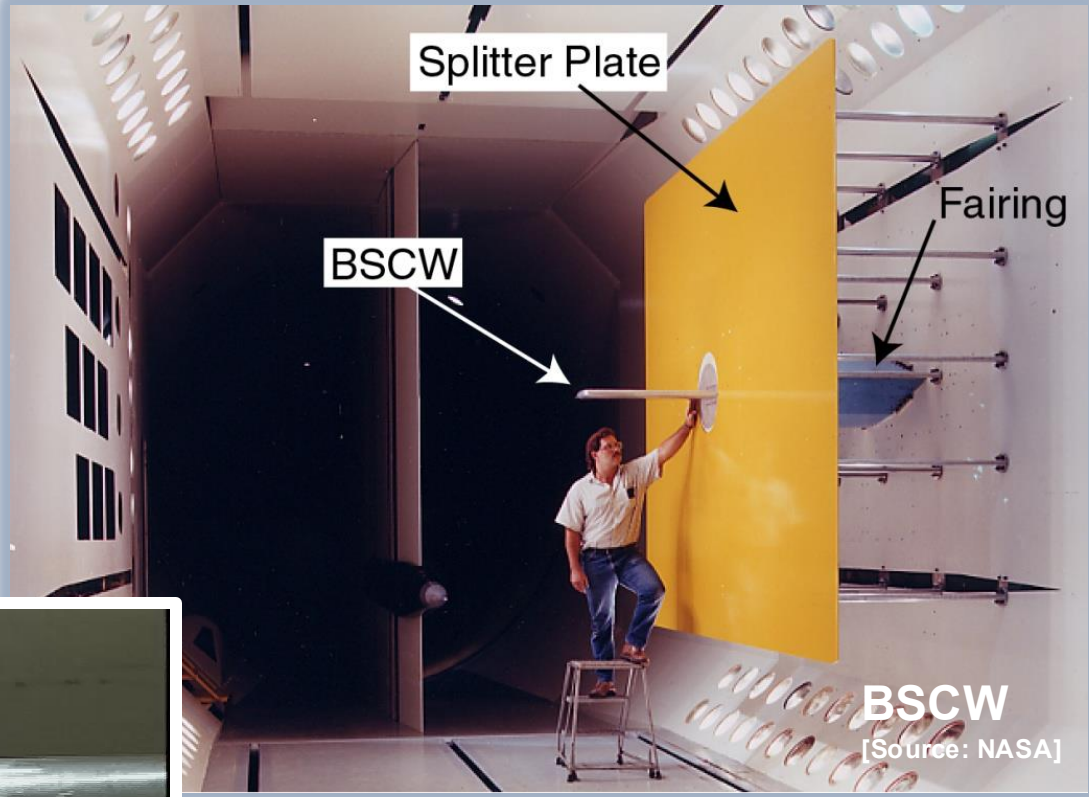
✓ - poor

✓ - mixed

AePW-3 Summary Papers: AIAA Paper 2024-0417 and 2024-0418

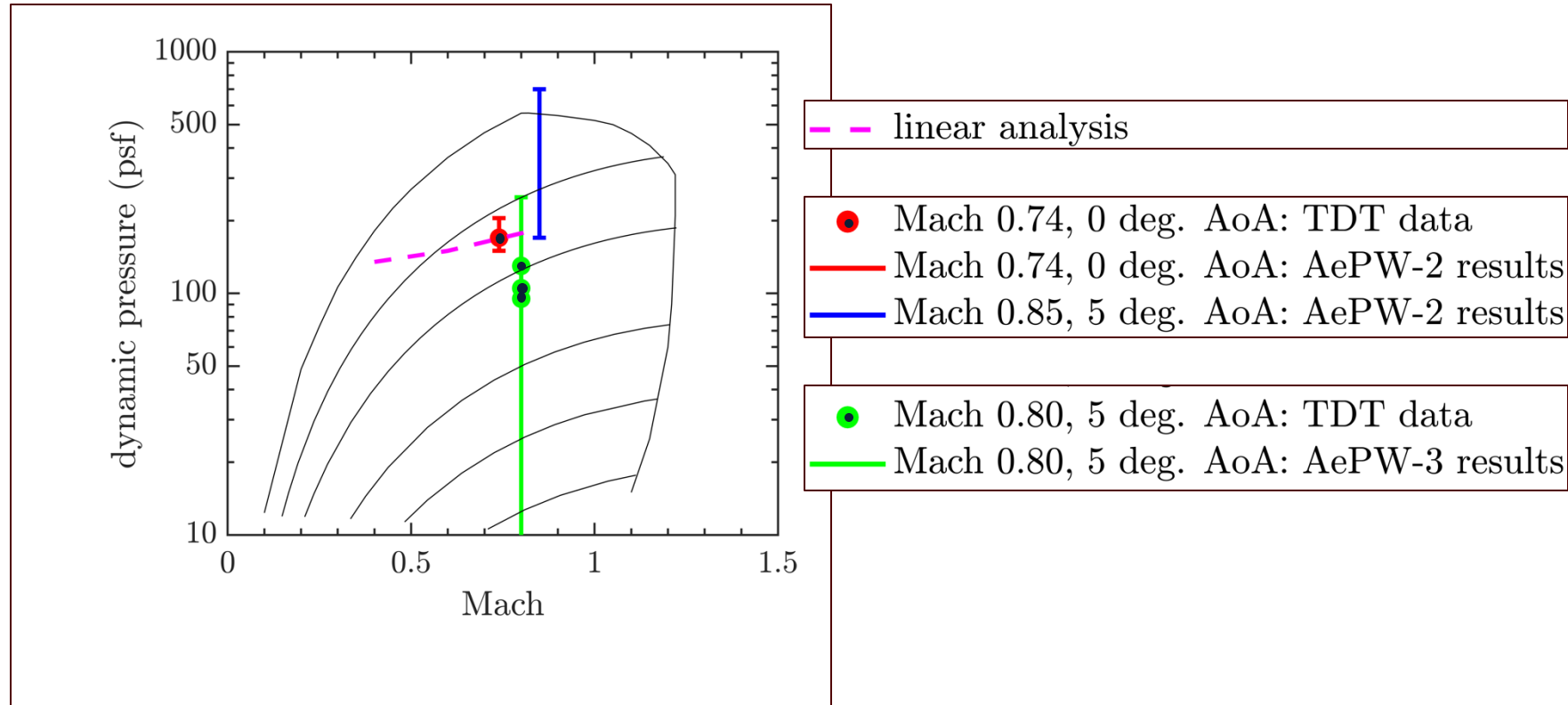
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BSCW Wing Configuration



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AePW-3: What have we learned?



- Large spread in BSCW flutter predictions from AePW-3 (though not as bad as AePW-2)
- We need more experimental data: more flutter data points, and more on-and off-body flow data at each flutter point

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BSCW wing configuration will be retested in TDT to obtain flutter and buffet data at Mach, dynamic pressure, and AoA range: September 2025

- Unsteady Pressure Sensitive Paint
- Flutter Stopper Safety Mechanism
- Two Rows of Pressure Sensors
 - + Several On Splitter Plate
- Particle Image Velocimetry, PIV

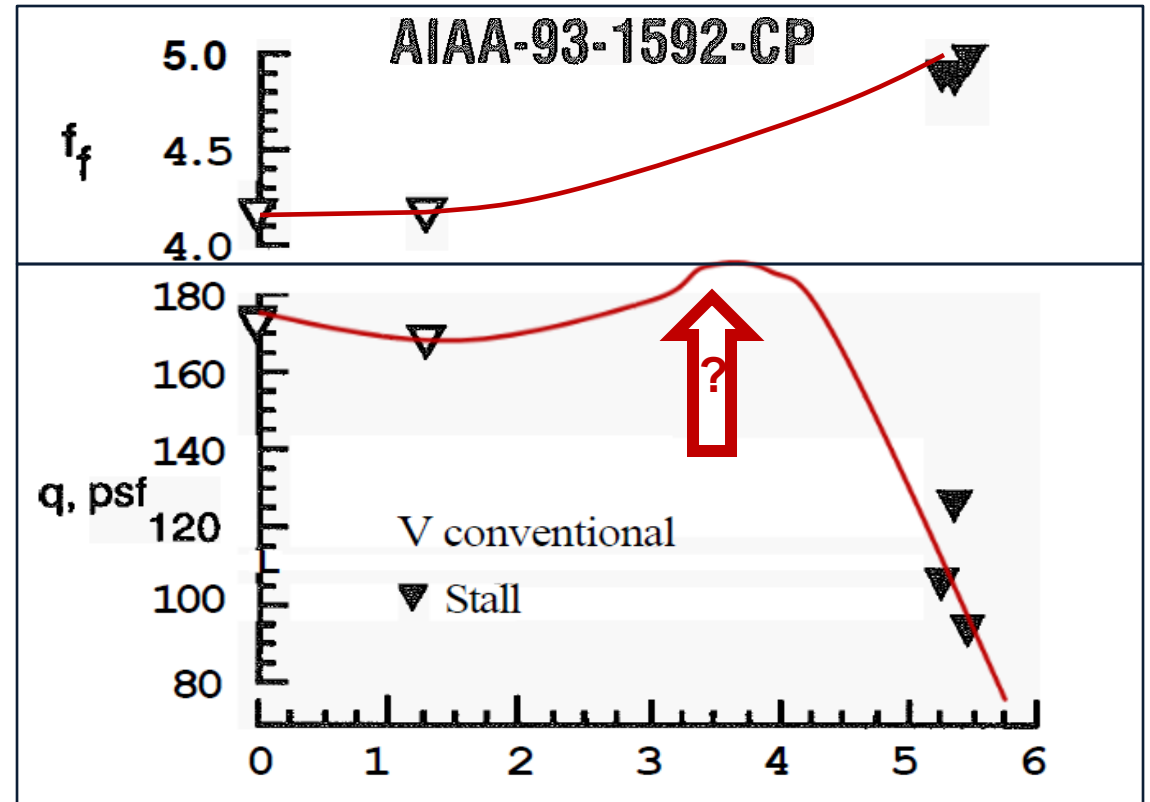


Figure 9. Stall flutter boundary in R-12 at $M = 0.80$.

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Workshop Cases:

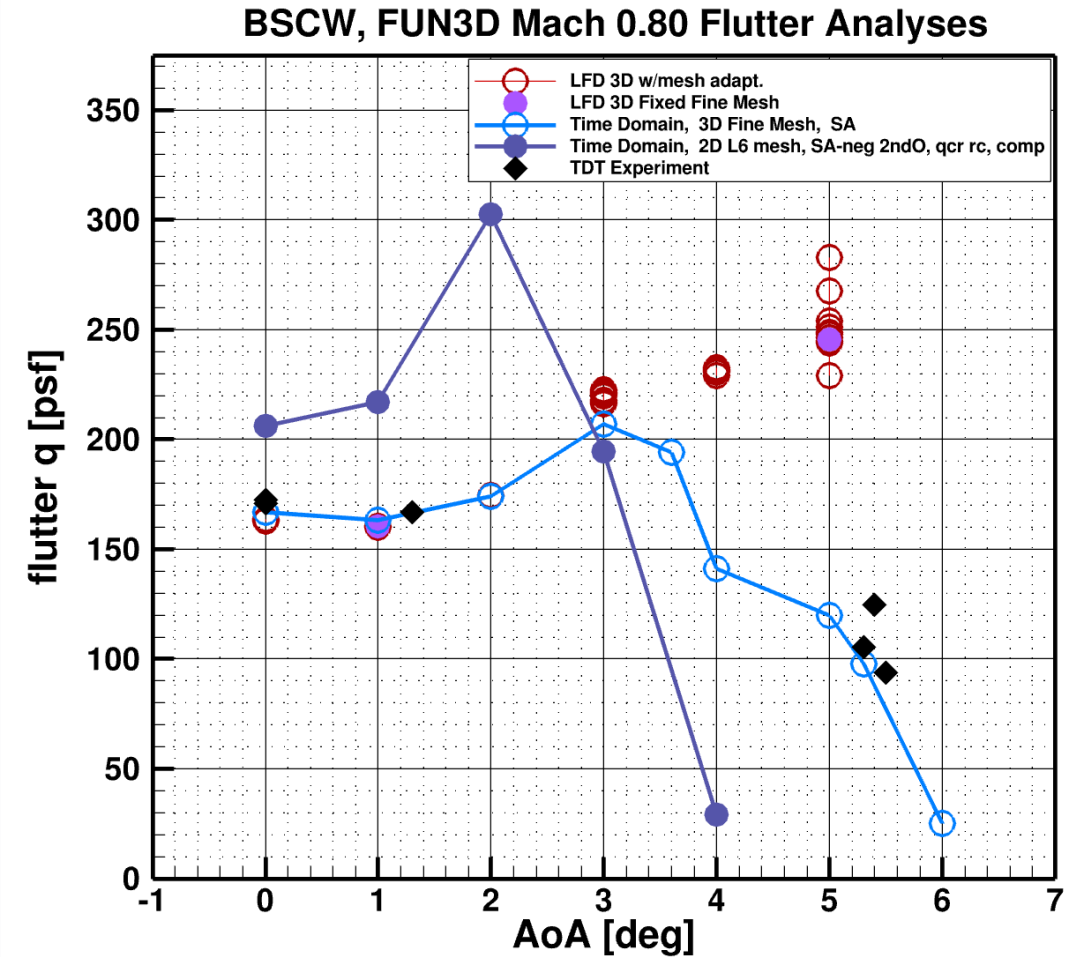
- **Mandatory**
 - **3D Flutter prediction at Mach 0.80 and angle-of-attack sweep: $0^{\circ} - 6^{\circ}$**
- **Optional**
 - **3D Flutter prediction at Mach 0.74, 0.76, 0.78 and angle-of-attack 3°**
- **Mandatory**
 - **2D BSCW flutter prediction at Mach 0.80 and angle-of-attack sweep: $0^{\circ} - 6^{\circ}$**

Currently we have seven teams looking at flutter calculations

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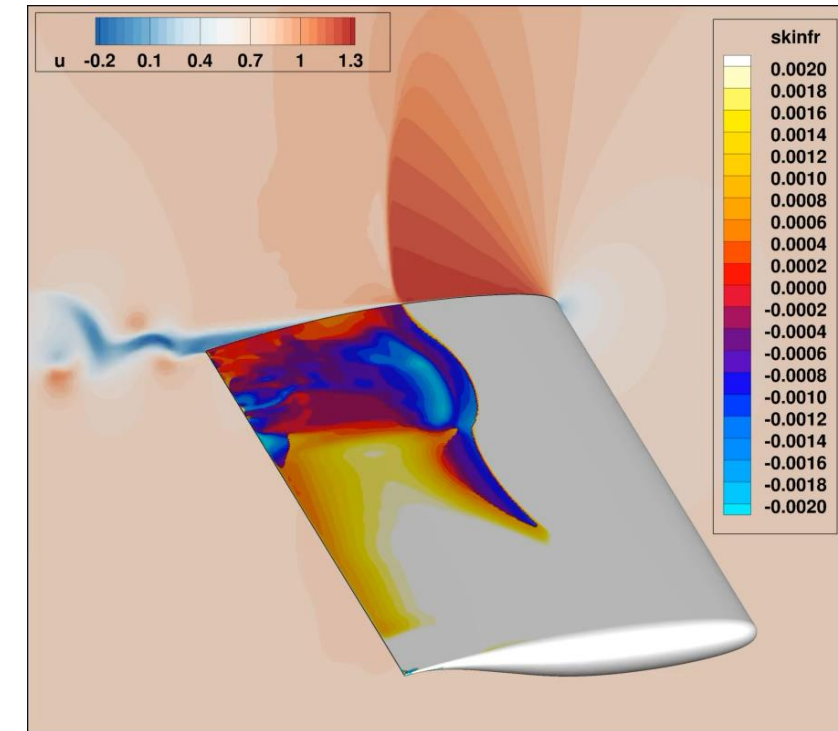
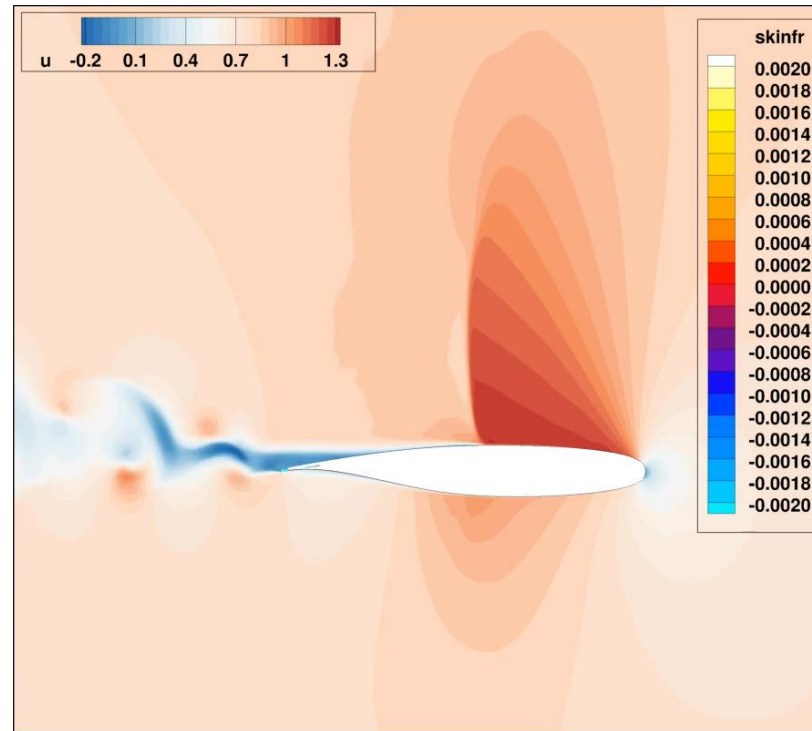
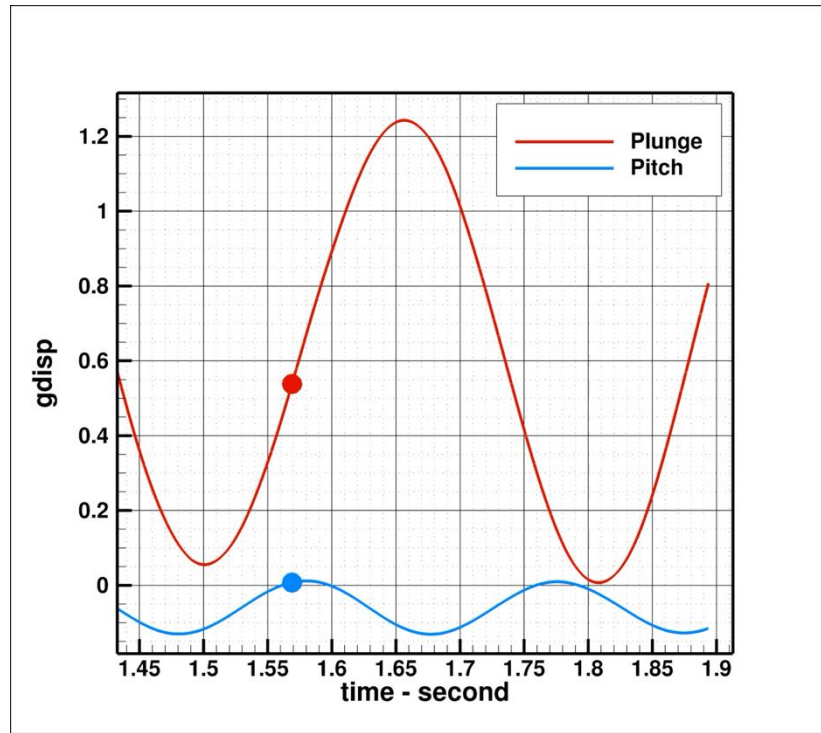
Sample of results from NASA Langley using FUN3D software:

- Linearized Frequency Domain and Time Domain methods
- 2D vs 3D
- Fixed mesh and with mesh adaptation



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- FUN3D DDES solution at Mach 0.8, 5deg AoA at $Q=100$ psf (near flutter), 99M mesh





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